Hybrid Neuro-Symbolic Machine Learning for Logical Reasoning Tasks: Sudoku Solver

Presented by Aakash and Nitish

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What is Neuro-Symbolic Machine Learning?

Bridging Two Worlds

Combines neural networks' learning power with symbolic logic' s reasoning and interpretability.

Synergistic AI

It bridges pattern recognition and rule-based problem-solving, creating a more robust AI.

Beyond Deep Learning

Enables AI to handle complex logical tasks that pure deep learning approaches often struggle with.

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Why Sudoku? A Logical Reasoning Benchmark

The Challenge

Sudoku is a 9x9 grid puzzle requiring precise logical deduction to fill missing numbers, making it an ideal test for AI.

The Perfect Testbed

It combines perception, constraints, and intricate reasoning. Traditional neural nets often falter with strict logical rules, where symbolic methods truly shine.

Neural Logic Machines (NLMs): The Hybrid Architecture

Pioneering Work

Introduced by Ashutosh Hathidara et al. (2023) specifically for Sudoku solving, setting a new standard.

Rule Encoding

Utilizes predicates that encode Sudoku rules as inputs to sophisticated neural-symbolic layers.

Constraint Satisfaction

Learns to infer missing numbers while strictly adhering to all Sudoku constraints, ensuring valid solutions.

High Accuracy

Achieves a remarkable 100% accuracy on puzzles with 3 to 10 empty cells, demonstrating its effectiveness.

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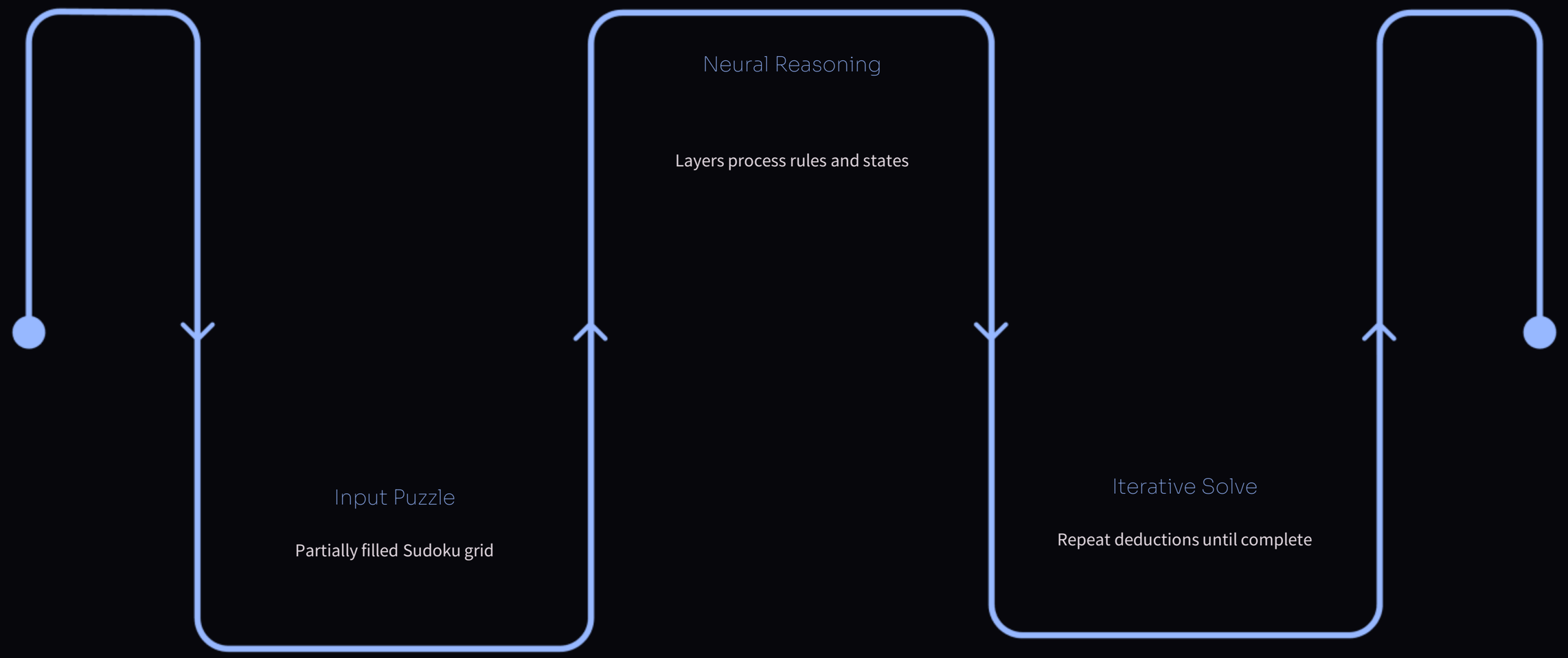
How NLMs Solve Sudoku: Step-by-Step



The NLM efficiently transforms a complex puzzle into a solvable format by iteratively applying learned rules and logical deductions.

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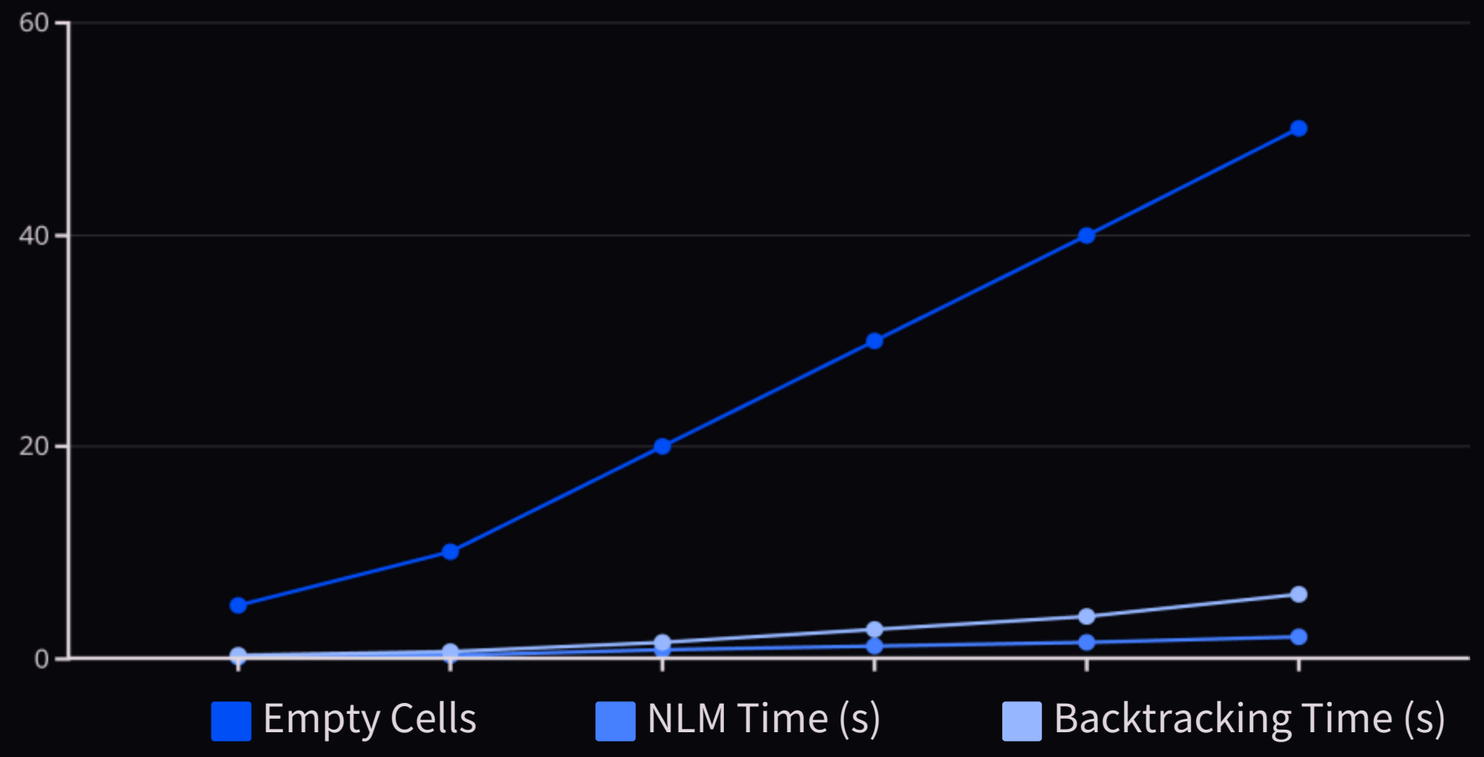
Performance vs Traditional Algorithms



Key Advantages

- NLMs trained with reinforcement learning often outperform standard deep nets on Sudoku.
- They achieve comparable accuracy to backtracking algorithms but with superior learned generalization.
- Faster convergence in complex puzzles is observed due to hybrid symbolic guidance.
- The chart illustrates NLM's efficiency, especially with increasing complexity.

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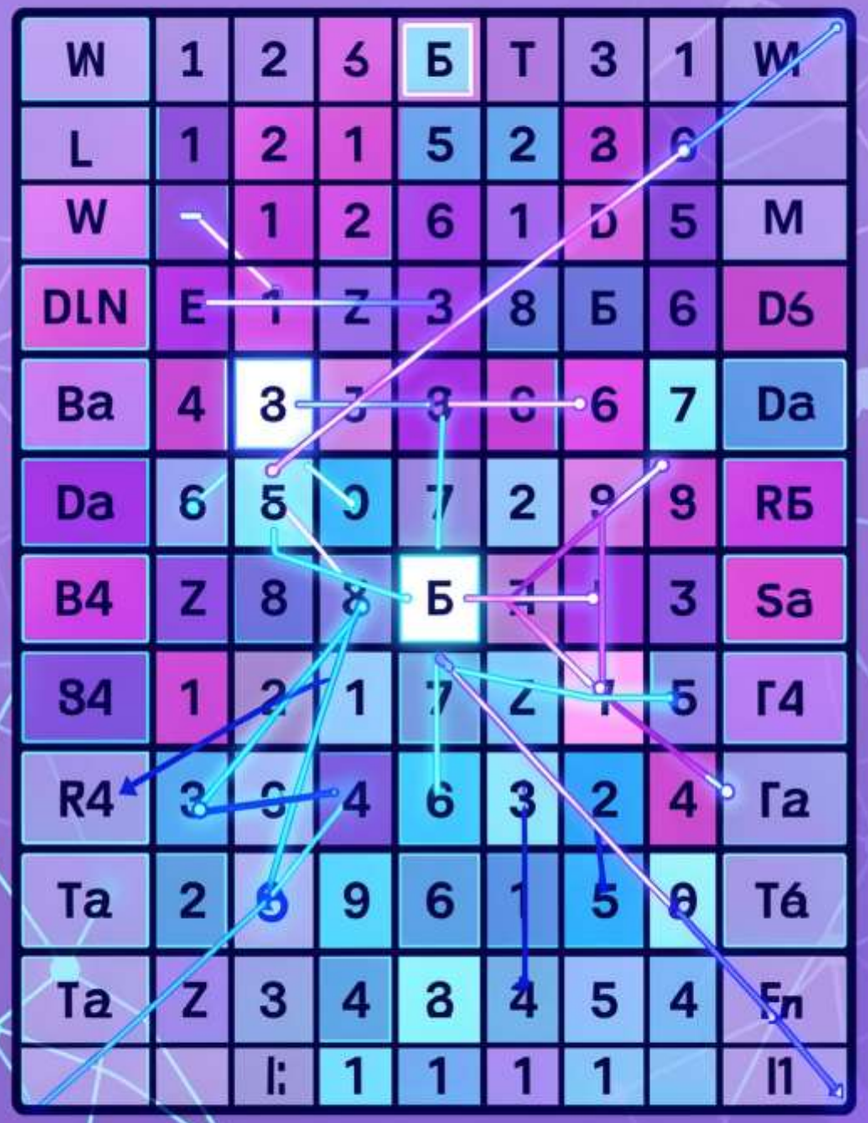

Visualizing Neuro-Symbolic Reasoning in Sudoku

From partial clues to logical completion, observe how the NLM deduces the correct numbers by applying a blend of learned patterns and hard-coded logical constraints, leading to a complete and valid Sudoku solution.

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Broader Impact: Beyond Sudoku

Enhanced Trustworthiness

Neuro-symbolic models improve AI trustworthiness by enforcing hard constraints, making their decisions more reliable.

Wide Applicability

Applicable to diverse fields such as theorem proving, program synthesis, and complex decision-making systems.

Minimal Data Learning

Recent advances, like the ASPER model, demonstrate promising results with significantly less training data.

Human-like Reasoning

This approach paves the way for AI systems that can learn and reason more intuitively, akin to human cognitive processes.

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Challenges and Future Directions

01

Scaling Complexity

Developing methods to scale these models to tackle even larger and more complex logical tasks.

02

Richer Knowledge Integration

Integrating more sophisticated symbolic knowledge into deep learning architectures for enhanced performance.

03

Improving Explainability

Focusing on enhancing the interpretability and explainability of decisions made by hybrid neuro-symbolic models.

04

Real-time Adaptation

Exploring the potential for real-time reasoning and adaptation in dynamic, unpredictable environments.

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Conclusion: The Future of Logical AI is Hybrid

- Neuro-symbolic learning seamlessly unites the strengths of both neural and symbolic worlds.
- The Sudoku solver stands as a testament to its practical success and remarkable generalization capabilities.
- Presented by Aakash and Nitish — we are dedicated to advancing AI reasoning, one puzzle at a time.

Thank you! Questions?